WORKED SOLUTION

STEP II 1988, Question 15

Probability and Statistics • Sums of Discrete Uniforms

STEP II 1988, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q15

The key idea

Each paper's mark is uniform on $\{0, 1, \ldots, m\}$, so every possible list of marks is equally likely, and a probability is just a count of lists divided by the total number of lists. The count for two papers comes from a simple listing, and the count for three papers comes from conditioning on the third mark and adding up two-paper counts (a *convolution*). Symmetry ($k \mapsto m - k$ on each paper) then gives the top range almost for free.

A word of warning before we start: the three-paper formula printed in the paper contains a misprint. We derive the correct one below and point out where the printed version differs.

Solution

Let X_1, X_2, X_3 be the marks on the papers. They are independent, since the papers are marked independently, and each is uniform on $\{0, 1, \ldots, m\}$. So each ordered pair (X_1, X_2) has probability $(m+1)^{-2}$, and each ordered triple has probability $(m+1)^{-3}$. Write $N_2(n)$ and $N_3(n)$ for the number of ordered pairs (triples) of marks in $\{0, \ldots, m\}$ with sum n.

Two papers

We need $X_1 = k$ and $X_2 = n - k$, with both in $[0, m]$, so

$$\max(0,\, n - m) \leq k \leq \min(m,\, n).$$

- If $0 \leq n \leq m$: k runs from 0 to n, so $N_2(n) = n + 1$.
- If $m < n \leq 2m$: k runs from $n - m$ to m, so $N_2(n) = m - (n - m) + 1 = 2m - n + 1$.

Hence

$$P(X_1 + X_2 = n) = \begin{cases} \dfrac{n+1}{(m+1)^2}, & 0 \le n \le m, \\ \dfrac{2m-n+1}{(m+1)^2}, & m < n \le 2m, \end{cases}$$

which is the required result for $m < n \le 2m$, together with the result for $0 \le n \le m$. The distribution is a "tent", rising by one each step up to $n = m$ and then falling.

Three papers

Condition on the third mark: if $X_3 = k$ then we need $X_1 + X_2 = n - k$, so

$$N_3(n) = \sum_{k=0}^{m} N_2(n-k) = \sum_{j=n-m}^{n} N_2(j),$$

where $j = n - k$ and $N_2(j) = 0$ when $j < 0$ or $j > 2m$.

Case $0 \le n \le m$**.** Only $j = 0, 1, \ldots, n$ contribute (negative j give zero), and all of them lie in the rising part of the tent:

$$N_3(n) = \sum_{j=0}^{n} (j+1) = \frac{(n+1)(n+2)}{2}, \qquad P(X_1 + X_2 + X_3 = n) = \frac{(n+1)(n+2)}{2(m+1)^3}.$$

Case $m < n \le 2m$**.** Now $1 \le n - m \le m < n \le 2m$, so the range $j = n - m, \ldots, n$ straddles the peak of the tent. We split it there:

$$N_3(n) = \sum_{j=n-m}^{m} (j+1) + \sum_{j=m+1}^{n} (2m - j + 1).$$

The first sum adds the integers from $n - m + 1$ to $m + 1$:

$$\sum_{j=n-m}^{m} (j+1) = \frac{(m+1)(m+2) - (n-m)(n-m+1)}{2}.$$

In the second sum, put $i = 2m - j + 1$, which runs from $2m - n + 1$ up to m:

$$\sum_{j=m+1}^{n} (2m - j + 1) = \frac{m(m+1) - (2m-n)(2m-n+1)}{2}.$$

Expanding the four products,

$$\begin{aligned} 2N_3(n) &= (m^2 + 3m + 2) - (n^2 - 2mn + m^2 + n - m) \\ &\quad + (m^2 + m) - (4m^2 - 4mn + n^2 + 2m - n) \\ &= 6mn - 3m^2 - 2n^2 + 3m + 2. \end{aligned}$$

Dividing by the $(m+1)^3$ equally likely triples,

$$P(X_1 + X_2 + X_3 = n) = \frac{6mn - 3m^2 - 2n^2 + 3m + 2}{2(m+1)^3}, \qquad m < n \le 2m.$$

Ask yourself This isn't the formula printed in the question. Who is right?

The printed version has $-4m^2$ in place of $-3m^2$ and $(m+1)^2$ in place of $(m+1)^3$; both are misprints. A single case settles it: with $m=1$ and $n=2$, three marks from $\{0,1\}$ add up to 2 in exactly 3 of the 8 equally likely ways, so the probability is $\frac{3}{8}$. Our formula gives $\frac{12-3-8+3+2}{16}=\frac{3}{8}$, while the printed one gives $\frac{12-4-8+3+2}{8}=\frac{5}{8}$. The denominator must be $(m+1)^3$ in any case, since that is the number of equally likely outcomes. As a further check, at $n=m$ our middle formula gives $\frac{(m+1)(m+2)}{2(m+1)^3}$, agreeing with the first case, as it should, since the two sums then join up.

Case $2m < n \le 3m$. Replace each mark X_i by $m - X_i$. This is again uniform on $\{0,\dots,m\}$, and the three new marks are still independent, so

$$\begin{aligned} P(X_1+X_2+X_3=n) &= P\big((m-X_1)+(m-X_2)+(m-X_3)=n\big) \\ &= P(X_1+X_2+X_3=3m-n). \end{aligned}$$

For $2m < n \le 3m$ we have $0 \le 3m-n < m$, so the first case applies with n replaced by $3m-n$:

$$P(X_1+X_2+X_3=n)=\frac{(3m-n+1)(3m-n+2)}{2(m+1)^3}, \qquad 2m<n\le 3m.$$

(The same symmetry, with $2m-n$ in place of $3m-n$, turns the two-paper result for $n \le m$ into the result for $n > m$.)

Checks

- Small case: for $m=1$ the three formulae give $\frac{1}{8},\frac{3}{8},\frac{3}{8},\frac{1}{8}$ for $n=0,1,2,3$, which add to 1 and match the binomial distribution of three fair coin tosses.
- The middle formula is symmetric under $n \mapsto 3m-n$: replacing n by $3m-n$ in $6mn-3m^2-2n^2$ leaves it unchanged, as the symmetry argument says it must.
- By complete enumeration for every m from 1 to 8, all five formulae agree exactly with the counts; for example, $m=10$ and $n=15$ give $\frac{91}{1331}$ both ways.

Common slips

- Forgetting that $N_2(j)$ has two different formulae, and using $j+1$ all the way through the sum in the middle range.
- Trying to reach the printed middle formula and "forcing" the algebra. If a careful calculation disagrees with a given answer, test a tiny case before doubting yourself.

Beyond STEP

The distribution of a sum of independent random variables is the *convolution* of their distributions, and that is exactly what our sum $\sum_k N_2(n-k)$ is. Probability generating functions make this mechanical: the count $N_3(n)$ is the coefficient of t^n in $(1+t+\cdots+t^m)^3$. With more and more papers, the shape goes from a tent to a smooth bump and approaches the normal curve (the central limit theorem). Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and the central limit theorem at the right level.

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